25900507 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT201 - BUILDING BLOCKS FOR HIGHER MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. What is meant by a direct proof of a conditional statement $p \to q$? [K] / [CO1]

2. Let $\mathcal{L}$ denote the set of all lines in a plane. For any two lines $l_1$and l_2,, we write if l_1 is perpendicular to l_2. Check whether ``perpendicular to'' is a relation on $\mathcal{L}$. [An] / [CO2]

3. Consider the poset $(\mathbf{Z} \times \mathbf{Z}, \preceq)$ where $\preceq$ is the lexicographic ordering constructed from the usual $\leq$ relation on $\mathbf{Z}$. Determine whether
   (a) $(3, 5) \prec (4, 8)$
   (b) $(3, 8) \prec (4, 5)$
   (c) $(4, 9) \prec (4, 11)$
   (d) $(5, 3) \prec (4, 4)$ [U] / [CO3]

4. Construct a partition on the set { a, b, c, e, f, b, i}. [U] / [CO3]

5. Find the unit tangent vector of the plane curve $\mathbf{r}(t) = t^2\mathbf{i} + e^t\mathbf{j}$. [A] / [CO4]

6. Find principal unit normal **N** for the space curve whose unit tangent vector **T** is given by $T = (\frac{1}{\sqrt{2}}\sin 4t)\mathbf{i} + (-\frac{1}{\sqrt{2}}\cos 4t)\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$. [A] / [CO4]

7. Test whether the differential form $(x+y)\,dx + y\,dy + (y+z)\,dz$ is exact. [K] / [CO5]

8. Find the curl of $\mathbf{F}(x,y,z) = xz\,\hat{i} + xy\,\hat{j} + 3xz\,\hat{k}$. [A] / [CO5]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. Prove that if x is irrational, then $\frac{1}{x}$ is irrational. [A] / [CO1]

10. (i) What do you mean by "circular reasoning"? [A] / [CO1]
(ii) Is the following argument correct? It supposedly shows that n is an even integer whenever n^2 is an even integer.
Suppose that n^2 is even. Then $n^2 = 2k$ for some integer k. Let $n = 2l$ for some integer l. This shows that n is even.

11. Determine whether the poset $(\mathcal{P}(S), \subseteq)$ is a lattice for any set S [U] / [CO3]

12. Draw the Hasse diagram representing the partial ordering {(a, b)|a divides b} on
{1, 2, 3, 4, 6, 8, 12}. [U] / [CO3]

13. Evaluate the speed and acceleration of the position vector
$r(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + \sin(2t)\,\mathbf{k}$ at $t = \frac{\pi}{2}$ [A] / [CO4]

14. Find ∇f if $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z + \tan^{-1} xz$, at (1, 1, 1). [U] / [CO4]

15. A fluid's velocity field is $\vec{F} = -4xy\mathbf{i} + 8y\mathbf{j} + 2\mathbf{k}$. Find the flow along the curve $\overrightarrow{r(t)} = t\mathbf{i} + t^2\mathbf{j} + \mathbf{k}, 0 \leq t \leq 2$. [An] / [CO5]

16. Find a potential function f for the field
$\mathbf{F} = 2xy\mathbf{i} + (x^2 - z^2)\mathbf{j} - 2yz\mathbf{k}$. [A] / [CO5]

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. Let R_1 and R_2 be relations on a set A represented by the matrices [An] / [CO2]
$M_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ and $M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Find the matrices

that represent a) $R_1 \cup R_2$. b) $R_1 \cap R_2$. c)$R_2 \circ R_1$. d) $R_1 \circ R_2$. e) $R_1 \oplus R_2$.

18. Show that $(\mathbb{Z}, \leq)$ is totally ordered, but not a well-ordered set. [U] / [CO3]

19. Let S a set of strings of zeros and ones. For any positive integer n , define the relation R_n on S such that sR_nt if and only if s = t , or both s and t have at least n characters and the first n characters of s and t are the same. Show that R_n is an equivalence relation. Also find three elements equivalent to 0101 in R_4. [U] / [CO3]

20. Consider the curve defined by [U] / [CO4]
$\mathbf{r}(t) = (1 + 2t)\mathbf{i} + (1 + 3t)\mathbf{j} + (6 - 6t)\mathbf{k}$
(a) Find the velocity vector and its magnitude.
(b) Find the arc length parameter $s(t)$ along the curve from the point where $t = 0$ by evaluating the integral $s = \int_0^t |\mathbf{v}(\tau)|\, d\tau$.
(c) Use the above arc length parameter to find the length of the curve for $-1 \leq t \leq 0$.

21. Find the line integral $\int_C (x^2 + y^2)\, dy$ along the path $C = C_1 \cup C_2$ consisting of the line segment from (0, 0) to (3, 0) followed by the segment from (3, 0) to (3, 3). [An] / [CO5]

22. Integrate $x + \sqrt{y}$ over the curve $C = C_1 \cup C_2$ where C_1 is the curve $y = x^2$ and C_2 is the straignt line $y = x$. [A] / [CO5]

[10x3 = 30]